Full Stack Assignment – 03

Assignment ID: FSD-03
Title: Internal Issue Tracking System
Duration: 3 Hours

1️⃣ Objective

Design and develop an **Internal Issue Tracking System** that allows employees to raise issues and administrators to manage their lifecycle.
The system must enforce status transitions and ensure **data persistence using a database**.

2️⃣ User Roles

🔷 EMPLOYEE

- Register and log in to the system
- Raise a new issue
- View own issues and their status

🔷 ADMIN

- Log in to the system
- View all reported issues
- Update issue status

3️⃣ Functional Requirements

Authentication & Authorization

- Implement user registration and login
- Use **JWT-based authentication**
- Protect all backend APIs using JWT
- Enforce **role-based access control** on APIs and frontend views

Issue Management

- EMPLOYEE can create issues with basic details
- Issues must be stored in the database
- ADMIN can update issue status
- Issue details must be retrieved from the database

4️⃣ Business Rule (Critical)

⚠️ **Issue status must follow this order only:**

OPEN → IN_PROGRESS → RESOLVED

Skipping steps or reverting to previous states is **not allowed**.

5️⃣ Technical Constraints (Mandatory)

- All issue data **must persist in a database**
- Hardcoded or in-memory issue storage is **not allowed**
- API responses must reflect real-time database state
- JWT authentication and role validation are mandatory
- Use environment variables for sensitive configuration

6️⃣ Frontend Requirements

- Login and registration pages
- Role-based dashboards for EMPLOYEE and ADMIN
- Issue creation form for EMPLOYEE
- Issue list view with current status
- Status update action for ADMIN

UI design is secondary; functional correctness is key.

7️⃣ Submission Requirements (Strict)

GitHub Repository

- Backend source code
- Frontend source code
- Proper folder structure
- Clear and meaningful commit history

Live Deployment

- Live frontend deployment link
- Live backend deployment link
- Backend must be connected to the database

README.md (Mandatory)

The README file must include:

- Project overview
- Tech stack used
- User roles and permissions
- API endpoints
- Database schema or collections

- Steps to run the project locally
- Live frontend and backend links

8️⃣ Evaluation Note

Evaluation focuses on **state management, role enforcement, and end-to-end correctness**, not on UI polish.